

Channel Estimation for Rydberg Atomic Receivers

Detailed Mathematical Reference for GD, Biased GS, PGD, CRLB, NMSE and Numerical Examples

Study and Simulation Reference

1 Purpose of This Document

This document is a detailed mathematical reference for understanding the paper “*Channel Estimation for Rydberg Atomic Receivers*”. It also uses the cited paper “*Towards Atomic MIMO Receivers*” to explain the biased Gerchberg–Saxton (GS) baseline.

The central problem is fundamentally different from ordinary wireless channel estimation.

In a conventional receiver, one may observe a linear model

$$Y = GS + N. \quad (1)$$

However, the Rydberg atomic receiver considered in the paper produces a magnitude-type measurement:

$$y_{i,p} = \left| \sum_{k=1}^K g_{i,k} s_{k,p} + b_{i,p} + n_{i,p} \right|. \quad (2)$$

Therefore, the complex phase information is hidden by the magnitude operation.

The conceptual progression of the paper is

Central Conceptual Transition

Complex multipath channel



Atomic magnitude-only observation



Strong reference signal



Taylor approximation



Approximately linear real-valued model



GD for 1D channel estimation



Low-rank 2D channel structure



2 Notation and Dimensions

Symbol	Dimension	Meaning
K	scalar	Number of users.
I	scalar	Number of vapor cells in a 1D array.
I_1, I_2	scalar	Dimensions of a 2D planar vapor-cell array.
P	scalar	Number of pilot time slots.
L_k	scalar	Number of propagation paths for user k .
$g_{i,k}$	complex scalar	Channel coefficient between user k and 1D cell i .
$g_{i_1, i_2, k}$	complex scalar	Channel coefficient between user k and 2D cell (i_1, i_2) .
G	$\mathbb{C}^{I \times K}$	1D channel matrix.
S	$\mathbb{C}^{K \times P}$	Pilot matrix.
B	complex matrix	Known reference signal.
Z	complex matrix	Reference phase compensation matrix.
Y	real matrix	Atomic receiver observations.
N	real matrix	Effective noise after approximation.
μ_{eg}	vector	Transition dipole moment vector.
$\alpha_{l,k}$	complex scalar	Gain of path l from user k .
$\epsilon_{i,k,l}$	vector	Polarization vector.
$\hbar$	scalar	Reduced Planck constant.
$\circ$	operator	Hadamard (element-wise) product.
H	superscript	Hermitian transpose.

3 Important Dimension Choice for Our Worked Examples

Throughout the numerical examples in this document, we use

$$\boxed{P = 3, \quad I = 3, \quad K = 3.} \quad (3)$$

Therefore, for the 1D case,

$$G \in \mathbb{C}^{3 \times 3}, \quad S \in \mathbb{C}^{3 \times 3}, \quad GS \in \mathbb{C}^{3 \times 3}. \quad (4)$$

This means we do **not** use a diagonal channel.

The channel contains all nine complex coefficients:

$$G = \begin{bmatrix} g_{1,1} & g_{1,2} & g_{1,3} \\ g_{2,1} & g_{2,2} & g_{2,3} \\ g_{3,1} & g_{3,2} & g_{3,3} \end{bmatrix}. \quad (5)$$

For the 2D example, we use

$$I_1 = 3, \quad I_2 = 3, \quad K = 3. \quad (6)$$

Hence each user's spatial channel is a full

$$3 \times 3 \quad (7)$$

complex matrix.

4 1D Rydberg Atomic Channel Model

The paper models the channel between the i -th vapor cell and the k -th user as

$$g_{i,k} = \sum_{l=1}^{L_k} \frac{1}{\hbar} \mu_{eg}^T \epsilon_{i,k,l} \alpha_{l,k} e^{-j(i-1)\varphi_{l,k}} \quad (7)$$

4.1 Meaning of Equation (7)

Each user does not necessarily reach the receiver through one propagation path.

Instead,

$$\text{Total channel} = \text{Path 1} + \text{Path 2} + \dots + \text{Path } L_k. \quad (8)$$

For every path:

- $\alpha_{l,k}$ determines complex path gain,
- $\epsilon_{i,k,l}$ describes polarization,
- $e^{-j(i-1)\varphi_{l,k}}$ describes spatial phase progression,
- $\mu_{eg}^T \epsilon_{i,k,l}$ describes atomic interaction with the incident electromagnetic polarization,
- $\hbar$ converts the physical interaction into the quantum coupling-scale coefficient.

4.2 Why the exponential appears

Suppose an electromagnetic wave reaches different vapor cells.

The wave reaches cell 1 first.

It reaches cell 2 with an additional phase delay.

It reaches cell 3 with another additional phase delay.

Therefore,

$$1, \quad e^{-j\varphi}, \quad e^{-j2\varphi}, \quad e^{-j3\varphi}, \dots \quad (9)$$

describe the spatial phase progression.

5 Atomic Magnitude Observation Model

For transmitted pilot $s_{k,p}$ from user k during pilot slot p , the received atomic measurement is

$$y_{i,p} = \left| \sum_{k=1}^K g_{i,k} s_{k,p} + b_{i,p} + n_{i,p} \right| \quad (8)$$

where

- $g_{i,k} s_{k,p}$ is the contribution of user k ,
- $b_{i,p}$ is the known reference signal,
- $n_{i,p}$ is complex noise,
- $|\cdot|$ destroys direct access to phase.

The known reference signal is

$$b_{i,p} = \frac{s_{b,p}}{\hbar} \mu_{eg}^T \epsilon_{b,i} \alpha_b e^{-j(i-1)\varphi_b} \quad (9)$$

This equation is important.

The denominator is the **reduced Planck constant**

$$\boxed{\hbar} \quad (10)$$

and not the ordinary Planck constant h .

6 Why Channel Estimation Is Difficult

Define

$$a_{i,p} = \sum_{k=1}^K g_{i,k} s_{k,p}. \quad (11)$$

Then Equation (8) becomes

$$y_{i,p} = |a_{i,p} + b_{i,p} + n_{i,p}|. \quad (12)$$

Suppose, for example,

$$x = 3 + 4j. \quad (13)$$

Then

$$|x| = 5. \quad (14)$$

But many different complex numbers have magnitude 5:

$$5, \quad 3 + 4j, \quad -3 + 4j, \quad -5j. \quad (15)$$

Therefore, knowing only the magnitude does not uniquely reveal the original complex number. This is the origin of the **phase retrieval problem**.

7 Strong Reference Approximation

The paper assumes the reference signal is much stronger than the desired signal and noise:

$$\boxed{|b_{i,p}| \gg |a_{i,p} + n_{i,p}|.} \quad (16)$$

Then

$$y_{i,p} = |a_{i,p} + b_{i,p} + n_{i,p}| \quad (17)$$

$$= |b_{i,p}| \sqrt{1 + \frac{|a_{i,p} + n_{i,p}|^2}{|b_{i,p}|^2} + 2 \operatorname{Re} \left(\frac{a_{i,p} + n_{i,p}}{b_{i,p}} \right)}. \quad (18)$$

Using

$$\sqrt{1+x} \approx 1 + \frac{x}{2}, \quad (19)$$

the nonlinear magnitude measurement is approximately converted into

$$\boxed{y_{i,p} = \operatorname{Re} \left(e^{-jz_{i,p}} \sum_{k=1}^K g_{i,k} s_{k,p} \right) + |b_{i,p}| + \bar{n}_{i,p}} \quad (11)$$

where

$$z_{i,p} = \angle b_{i,p}. \quad (20)$$

The effective noise is

$$\bar{n}_{i,p} = \operatorname{Re} \left(e^{-j\angle b_{i,p}} n_{i,p} \right). \quad (21)$$

If

$$n_{i,p} \sim \mathcal{CN}(0, \sigma^2), \quad (22)$$

then

$$\bar{n}_{i,p} \sim \mathcal{N} \left(0, \frac{\sigma^2}{2} \right). \quad (23)$$

8 Matrix Form of the 1D Observation Model

Define

$$G \in \mathbb{C}^{I \times K}, \quad S \in \mathbb{C}^{K \times P}. \quad (24)$$

Then

$$GS \in \mathbb{C}^{I \times P}. \quad (25)$$

Define

$$Z_{i,p} = e^{-jz_{i,p}}. \quad (26)$$

The complete matrix model becomes

$$\boxed{Y = \text{Re}[Z \circ (GS)] + |B| + N} \quad (12)$$

This is the central equation used by the GD method.

9 Gradient Descent (GD)

9.1 Basic idea

Suppose you are standing on a mountain in fog.

You cannot see the bottom.

But you can check the slope beneath your feet.

You take a small step downhill.

Then you check again.

Gradient descent does exactly this mathematically.

The objective function in the paper is

$$\boxed{\min_G L(G) = \|Y - |B| - \text{Re}[Z \circ (GS)]\|_F^2} \quad (13)$$

The term inside the norm is the **residual**:

$$R = Y - |B| - \text{Re}[Z \circ (GS)]. \quad (27)$$

GD tries to make this residual as small as possible.

9.2 GD Update Equation

The update rule is

$$\boxed{G^{(t+1)} = G^{(t)} - \eta_t \nabla_{G^*} L(G^{(t)})} \quad (14)$$

where η_t is the step size.

The gradient is

$$\nabla_G L(G^{(t)}) = -2 \left(\left[Y - |B| - \text{Re} \left(Z \circ (G^{(t)} S) \right) \right] \circ Z^* \right) S^H \quad (15)$$

9.3 What Equation (15) is doing

Equation (15) has three major operations:

1. Calculate prediction error,
2. Rotate the error according to reference phase,
3. Map the error back through the pilot matrix.

In words:

$$\text{Prediction} \rightarrow \text{Error} \rightarrow \text{Gradient} \rightarrow \text{Update channel} \quad (28)$$

10 Full Numerical Example: GD with Complex Matrices

We now solve one actual GD iteration.

We use

$$I = 3, \quad K = 3, \quad P = 3. \quad (29)$$

The following complex matrices are educational numerical realizations. They are deliberately non-diagonal and nontrivial.

10.1 Step 1: True Channel Matrix

Let

$$G_{\text{true}} = \begin{bmatrix} 0.82 + 0.31j & -0.47 + 0.68j & 1.13 - 0.26j \\ -0.35 + 0.91j & 0.74 - 0.52j & -0.63 + 0.44j \\ 1.05 + 0.17j & -0.28 - 0.76j & 0.56 + 0.83j \end{bmatrix}. \quad (30)$$

This is a full complex 3×3 channel.

For example,

$$g_{1,2} = -0.47 + 0.68j \quad (31)$$

means the channel between user 2 and vapor cell 1 has

- amplitude information,
- phase information.

10.2 Step 2: Complex Pilot Matrix

Let

$$S = \begin{bmatrix} 0.71 + 0.42j & -0.63 + 0.81j & 0.35 - 0.94j \\ -0.52 + 0.73j & 0.88 + 0.21j & -0.77 - 0.46j \\ 0.93 - 0.18j & -0.31 + 0.67j & 0.49 + 0.58j \end{bmatrix}. \quad (32)$$

Rows correspond to users.

Columns correspond to pilot time slots.

Therefore:

$$S = \begin{bmatrix} \text{User 1 pilots} \\ \text{User 2 pilots} \\ \text{User 3 pilots} \end{bmatrix}. \quad (33)$$

10.3 Step 3: Calculate the Complex Signal $A = G_{\text{true}}S$

Multiplying the matrices gives

$$A = G_{\text{true}}S \quad (34)$$

and numerically

$$A = \begin{bmatrix} 1.2041 - 0.5774j & -1.5002 + 1.8063j & 1.9576 - 0.4417j \\ -1.1426 + 1.8323j & 0.1443 - 1.7175j & -0.6400 + 0.5577j \\ 2.0447 + 1.4236j & -1.6157 + 0.1337j & 0.1863 + 0.5180j \end{bmatrix}. \quad (35)$$

Consider the first element:

$$a_{1,1} = g_{1,1}s_{1,1} + g_{1,2}s_{2,1} + g_{1,3}s_{3,1} \quad (36)$$

$$= (0.82 + 0.31j)(0.71 + 0.42j) \quad (37)$$

$$+ (-0.47 + 0.68j)(-0.52 + 0.73j) \quad (38)$$

$$+ (1.13 - 0.26j)(0.93 - 0.18j) \quad (39)$$

$$= 1.2041 - 0.5774j. \quad (40)$$

This is actual complex multi-user signal superposition.

10.4 Step 4: Reference Signal

For the educational example, let the reference have magnitude 5 with phase matrix

$$\Phi_B = \begin{bmatrix} 0.2 & 1.0 & -0.6 \\ 0.7 & -1.1 & 0.4 \\ -0.9 & 0.5 & 1.2 \end{bmatrix}. \quad (41)$$

Then

$$B = 5e^{j\Phi_B}. \quad (42)$$

The phase compensation matrix is

$$Z = e^{-j\Phi_B}. \quad (43)$$

Since the reference magnitude is 5,

$$|B| = 5 \begin{bmatrix} 1 & 1 & 1 \\ 1 & 1 & 1 \\ 1 & 1 & 1 \end{bmatrix}. \quad (44)$$

10.5 Step 5: Generate the Linearized Observation

Ignoring noise for this worked iteration,

$$Y = |B| + \text{Re} [Z \circ A]. \quad (45)$$

Numerically,

$$Y = \begin{bmatrix} 6.0654 & 5.7094 & 6.8651 \\ 5.3065 & 6.5961 & 4.6277 \\ 5.1559 & 3.6462 & 5.5503 \end{bmatrix}. \quad (46)$$

Notice that Y is real.

That is important.

The original channel is complex, but after the strong-reference approximation the measurement model contains the real part.

10.6 Step 6: Start GD from a Wrong Initial Guess

Suppose GD starts from

$$G^{(0)} = \begin{bmatrix} 0.15 - 0.10j & -0.20 + 0.05j & 0.10 + 0.12j \\ 0.05 + 0.08j & -0.12 + 0.15j & 0.18 - 0.06j \\ -0.07 + 0.11j & 0.09 - 0.14j & -0.16 + 0.04j \end{bmatrix}. \quad (47)$$

Clearly,

$$G^{(0)} \neq G_{\text{true}}. \quad (48)$$

GD must gradually improve this estimate.

10.7 Step 7: Current Prediction

First calculate

$$\hat{Y}^{(0)} = \text{Re} [Z \circ (G^{(0)} S)]. \quad (49)$$

Numerically,

$$\hat{Y}^{(0)} = \begin{bmatrix} 0.3068 & 0.0138 & 0.0980 \\ -0.0282 & -0.3234 & 0.3457 \\ -0.3115 & -0.0919 & 0.0342 \end{bmatrix}. \quad (50)$$

10.8 Step 8: Calculate the Residual

The residual is

$$R^{(0)} = Y - |B| - \hat{Y}^{(0)}. \quad (51)$$

Numerically,

$$R^{(0)} = \begin{bmatrix} 0.7585 & 0.6956 & 1.7670 \\ 0.3347 & 1.9195 & -0.7180 \\ 0.4674 & -1.2619 & 0.5161 \end{bmatrix}. \quad (52)$$

Interpretation:

- Positive value means the model predicts too little.
- Negative value means the model predicts too much.
- Large magnitude means a larger mismatch.

10.9 Step 9: Calculate the Gradient

Using Equation (15),

$$\nabla_G L(G^{(0)}) = -2 \left(R^{(0)} \circ Z^* \right) S^H. \quad (53)$$

The numerical gradient is

$$\nabla_G L(G^{(0)}) = \begin{bmatrix} -4.5536 - 0.2865j & 0.9738 - 2.5085j & -2.1517 + 2.9881j \\ 3.2610 + 0.6028j & -2.1381 + 4.1522j & 3.4060 - 0.8802j \\ 0.2531 - 2.4806j & 3.7704 + 1.2118j & -1.2894 - 1.5371j \end{bmatrix}. \quad (54)$$

This matrix tells GD how every channel coefficient should change.

For example, the gradient corresponding to $g_{1,1}$ is

$$-4.5536 - 0.2865j. \quad (55)$$

Therefore GD knows that both real and imaginary parts require correction.

10.10 Step 10: Perform One GD Update

Choose

$$\eta = 0.05. \quad (56)$$

Then

$$G^{(1)} = G^{(0)} - 0.05 \nabla_G L(G^{(0)}). \quad (57)$$

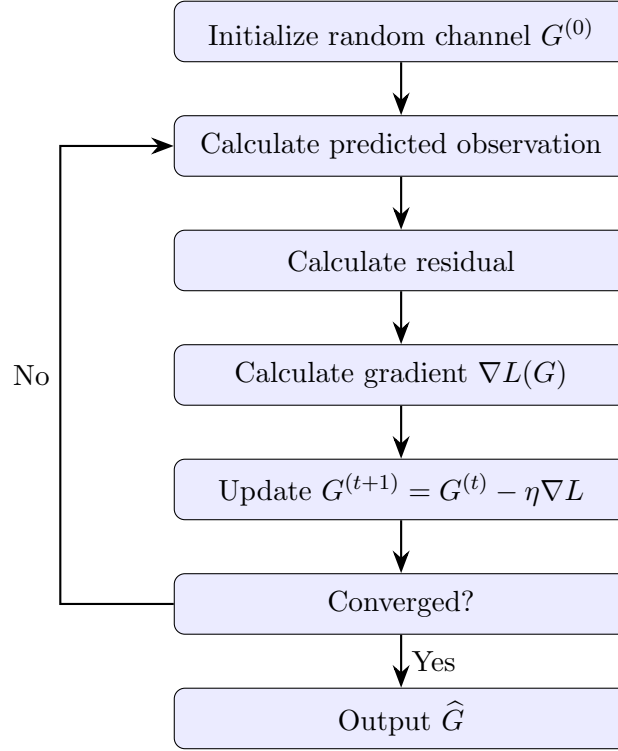
Numerically,

$$G^{(1)} = \begin{bmatrix} 0.3777 - 0.0857j & -0.2487 + 0.1754j & 0.2076 - 0.0294j \\ -0.1130 + 0.0499j & -0.0131 - 0.0576j & 0.0097 - 0.0160j \\ -0.0827 + 0.2340j & -0.0985 - 0.2006j & -0.0955 + 0.1169j \end{bmatrix}. \quad (58)$$

Then the process repeats:

$$G^{(0)} \rightarrow G^{(1)} \rightarrow G^{(2)} \rightarrow \dots \rightarrow \hat{G}. \quad (59)$$

11 GD Algorithm Flow



12 Gerchberg–Saxton (GS) Algorithm

The GS algorithm comes from phase retrieval.

The fundamental problem is

$$z = |A^H s + b + w|. \quad (60)$$

If phase were known, then

$$z \circ e^{j\theta} = A^H s + b + w. \quad (61)$$

Then channel or symbol recovery would become a normal linear least-squares problem.

The difficulty is:

$$\boxed{\text{Magnitude is known, but phase is unknown.}} \quad (62)$$

GS repeatedly guesses and improves the missing phase.

12.1 Biased GS Equations

The biased GS algorithm used in *Towards Atomic MIMO Receivers* works with the reference signal explicitly.

The phase update is

$$\boxed{\theta^{(t)} = \angle \left(A^H s^{(t-1)} + b \right)} \quad (63)$$

Then the least-squares update is

$$\boxed{s^{(t)} = (AA^H)^{-1} A \left(z \circ e^{j\theta^{(t)}} - b \right)} \quad (64)$$

The idea is simple:

1. Guess phase,
2. Restore a complex-valued signal using measured magnitude,
3. Remove the known reference,
4. Solve a linear regression problem,
5. Obtain a better estimate,
6. Calculate a new phase,
7. Repeat.

13 How GS Becomes a Channel Estimator

For channel estimation at one receive element, the cited paper writes the pilot observation in the form

$$z_n = |S^H a_n + b_n + w_n|. \quad (65)$$

Here:

- a_n is the unknown channel vector,
- S is the known pilot matrix,
- b_n is the known reference,
- z_n contains magnitude measurements.

Thus, for channel estimation we can identify

$$A = S. \quad (66)$$

The GS iteration becomes

$$\theta_n^{(t)} = \angle \left(S^H a_n^{(t-1)} + b_n \right) \quad (67)$$

followed by

$$a_n^{(t)} = (SS^H)^{-1} S \left[z_n \circ e^{j\theta_n^{(t)}} - b_n \right]. \quad (68)$$

The procedure is repeated independently for each receive antenna or vapor cell.

14 Full Numerical Example: GS Core Iteration

We use the same 3×3 complex system.

The exact nonlinear magnitude observation is

$$Y_{\text{NL}} = |G_{\text{true}}S + B|. \quad (69)$$

Numerically,

$$Y_{\text{NL}} = \begin{bmatrix} 6.1186 & 6.1325 & 6.9049 \\ 5.7208 & 6.6281 & 4.6902 \\ 5.7242 & 3.7537 & 5.5503 \end{bmatrix}. \quad (70)$$

Unlike GD, GS works directly with this nonlinear magnitude measurement.

14.1 One Cell / Channel-Vector View

Consider vapor cell 1.

Its true channel vector is

$$a_1 = \begin{bmatrix} 0.82 + 0.31j \\ -0.47 + 0.68j \\ 1.13 - 0.26j \end{bmatrix}. \quad (71)$$

The three pilot observations are

$$z_1 = \begin{bmatrix} 6.1186 \\ 6.1325 \\ 6.9049 \end{bmatrix}. \quad (72)$$

The reference vector is the first row of B :

$$b_1 = 5 \begin{bmatrix} e^{j0.2} \\ e^{j1.0} \\ e^{-j0.6} \end{bmatrix}. \quad (73)$$

14.2 GS Step 1: Start with a Channel Guess

Suppose

$$a_1^{(0)} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}. \quad (74)$$

Then the predicted complex received vector is initially dominated by the reference:

$$S^H a_1^{(0)} + b_1 = b_1. \quad (75)$$

Therefore the initial phase estimate is

$$\theta_1^{(1)} = \angle b_1. \quad (76)$$

14.3 GS Step 2: Restore Complex Measurement

GS knows the measured magnitude z_1 .

It attaches the estimated phase:

$$\tilde{y}_1 = z_1 \circ e^{j\theta_1^{(1)}}. \quad (77)$$

Then remove the known reference:

$$q_1 = z_1 \circ e^{j\theta_1^{(1)}} - b_1. \quad (78)$$

Doing this for all three vapor cells gives the matrix

$$Q^{(1)} = \begin{bmatrix} 1.0963 + 0.2222j & 0.6119 + 0.9529j & 1.5722 - 1.0756j \\ 0.5513 + 0.4644j & 0.7385 - 1.4510j & -0.2854 - 0.1207j \\ 0.4501 - 0.5673j & -1.0937 - 0.5975j & 0.1994 + 0.5129j \end{bmatrix}. \quad (79)$$

14.4 GS Step 3: Linear Regression Projection

Now solve

$$G_{\text{GS}}^{(1)} = Q^{(1)} S^H (S S^H)^{-1}. \quad (80)$$

Numerically,

$$G_{\text{GS}}^{(1)} = \begin{bmatrix} 1.2141 + 0.7822j & 0.8453 - 0.9183j & 0.7328 - 1.9416j \\ 0.5363 + 0.1622j & -0.0482 - 1.6618j & -0.8945 - 0.9312j \\ -0.0171 - 0.2688j & -1.2821 - 0.0226j & -0.4572 + 0.5081j \end{bmatrix}. \quad (81)$$

This is not yet the true channel.

It is simply the first phase-retrieval estimate.

Then GS repeats:

$$G^{(1)} \rightarrow \theta^{(2)} \rightarrow G^{(2)} \rightarrow \theta^{(3)} \rightarrow G^{(3)} \rightarrow \dots \quad (82)$$

15 Important Difference Between GS and GD

Feature	GS	GD
Starting model	Nonlinear magnitude model	Strong-reference linearized model
Main unknown	Missing phase	Channel coefficients
Main operation	Alternate phase restoration and LS projection	Move downhill using gradient
Reference signal	Explicitly removed after phase restoration	Used to linearize the observation
Noise treatment	Basic GS LS version does not explicitly model noise distribution	Least-squares optimization after approximation
2D low-rank constraint	Not naturally incorporated	Can be extended using projection
2D extension	Difficult	PGD handles it naturally

A simple memory rule is:

GS asks: “What phase should I attach to the measured magnitude?”

GD asks: “How should I change the channel coefficients to reduce the prediction error?”

16 Extension from 1D to 2D Antenna Arrays

Now suppose vapor cells are arranged in a planar array.

Instead of one index i , each cell has two indices:

$$(i_1, i_2). \quad (83)$$

The array size is

$$I_1 \times I_2. \quad (84)$$

The paper’s 2D channel equation is

$$g_{i_1, i_2, k} = \sum_{l=1}^{L_k} \frac{1}{\hbar} \mu_{eg}^T \epsilon_{i_1, i_2, k, l} \alpha_{l, k} e^{-j[(i_1-1)u_{l, k} + (i_2-1)v_{l, k}]} \quad (16)$$

Again, the denominator is

$$\boxed{\hbar}. \quad (85)$$

16.1 Meaning of $u_{l, k}$ and $v_{l, k}$

The paper defines

$$u_{l,k} = \frac{2\pi d_1}{\lambda} \cos \theta_{l,k}, \quad (86)$$

and

$$v_{l,k} = \frac{2\pi d_2}{\lambda} \sin \theta_{l,k} \cos \phi_{l,k}. \quad (87)$$

The variables

- $u_{l,k}$ control phase progression along dimension 1,
- $v_{l,k}$ control phase progression along dimension 2.

Therefore the 2D array captures spatial variation in two directions.

17 2D Reference Signal

The paper defines

$$b_{i_1, i_2, p} = \frac{s_{b,p}}{\hbar} \mu_{eg}^T \epsilon_{b, i_1, i_2} \alpha_b e^{-j[(i_1-1)u_b + (i_2-1)v_b]} \quad (17)$$

Again, the known reference signal has spatial phase progression in both dimensions.

18 2D Observation Model

After the strong-reference approximation,

$$y_{i_1, i_2, p} = \text{Re} \left(e^{-jz_{i_1, i_2, p}} \sum_{k=1}^K g_{i_1, i_2, k} s_{k, p} \right) + |b_{i_1, i_2, p}| + n_{i_1, i_2, p} \quad (18)$$

The complete tensor model is

$$\mathcal{Y} = \text{Re} [\mathcal{Z} \circ (\mathcal{G} \times_3 S^T)] + |\mathcal{B}| + \mathcal{N} \quad (19)$$

where

$$\mathcal{G} \in \mathbb{C}^{I_1 \times I_2 \times K}. \quad (88)$$

19 What Is a Tensor Here?

A vector has one dimension:

$$[x_1, x_2, x_3]. \quad (89)$$

A matrix has two dimensions:

$$\begin{bmatrix} x_{11} & x_{12} \\ x_{21} & x_{22} \end{bmatrix}. \quad (90)$$

A tensor here has three dimensions:

$$\mathcal{G} \in \mathbb{C}^{I_1 \times I_2 \times K}. \quad (91)$$

For every user k , one frontal slice is

$$G_k = \mathcal{G}(:, :, k) \in \mathbb{C}^{I_1 \times I_2}. \quad (92)$$

Thus:

$$\boxed{\text{User 1 channel} = G_1} \quad (93)$$

$$\boxed{\text{User 2 channel} = G_2} \quad (94)$$

$$\boxed{\text{User 3 channel} = G_3}. \quad (95)$$

20 Why the 2D Channel Is Low Rank

Consider one propagation path.

Its spatial structure is

$$e^{-j[(i_1-1)u+(i_2-1)v]}. \quad (96)$$

This can be separated:

$$e^{-j[(i_1-1)u+(i_2-1)v]} = e^{-j(i_1-1)u} e^{-j(i_2-1)v}. \quad (97)$$

Therefore one path can be represented approximately as

$$\mathbf{a}(u)\mathbf{b}(v)^T. \quad (98)$$

This is an outer product.

An outer product has rank one.

If there are L_k paths,

$$G_k = \sum_{l=1}^{L_k} \text{rank-one matrix}. \quad (99)$$

Therefore,

$$\boxed{\text{rank}(G_k) \leq L_k}. \quad (100)$$

This is the physical reason PGD works.

21 Mode-3 Unfolding

The paper unfolds the tensor along mode 3.

The unfolded model is

$$\boxed{Y_{(3)} = \text{Re} [Z_{(3)} \circ (S^T G_{(3)})] + |B_{(3)}| + N_{(3)}} \quad (20)$$

The unfolded channel matrix has dimension

$$G_{(3)} \in \mathbb{C}^{K \times I_1 I_2}. \quad (101)$$

For our example:

$$K = 3, \quad I_1 = 3, \quad I_2 = 3. \quad (102)$$

Therefore,

$$G_{(3)} \in \mathbb{C}^{3 \times 9}. \quad (103)$$

Each row contains the vectorized 3×3 channel matrix of one user.

22 2D Rank-Constrained Optimization

The optimization problem is

$$\boxed{\begin{array}{ll} \min_{G_{(3)}} & \|Y_{(3)} - |B_{(3)}| - \text{Re} [Z_{(3)} \circ (S^T G_{(3)})]\|_F^2 \\ \text{subject to} & \text{rank}(\text{mat}([G_{(3)}]_{k,:})) \leq L_k, \\ & k = 1, 2, \dots, K. \end{array}} \quad (21)$$

This is the key difference between 1D and 2D.

In 1D:

$$\text{Estimate channel.} \quad (104)$$

In 2D:

$$\text{Estimate channel + enforce physical low-rank structure.} \quad (105)$$

23 Projected Gradient Descent (PGD)

PGD consists of two operations.

23.1 Operation 1: Gradient Descent

Move toward lower estimation error:

$$M = G_{(3)}^{(t)} - \zeta_t \nabla f(G_{(3)}^{(t)}). \quad (106)$$

However, this matrix M may violate the rank constraint.

Therefore we need a second operation.

23.2 Operation 2: Projection

Project M onto the physically valid low-rank set.

Thus:

$$\boxed{\text{PGD} = \text{Gradient Step} + \text{Low-rank Projection.}} \quad (107)$$

24 PGD Objective Function

The paper defines

$$\boxed{f(G_{(3)}) = \|Y_{(3)} - |B_{(3)}| - \text{Re}[Z_{(3)} \circ (S^T G_{(3)})]\|_F^2} \quad (22)$$

The gradient used by the PGD procedure is

$$\boxed{\nabla_{G_{(3)}} f(G_{(3)}^{(t)}) = -S^* \left(\left[Y_{(3)} - |B_{(3)}| - \text{Re} \left(Z_{(3)} \circ (S^T G_{(3)}^{(t)}) \right) \right] \circ Z^* \right)} \quad (24)$$

25 Projection Problem

After a gradient step, PGD finds the closest valid low-rank matrix:

$$\boxed{\text{proj}_{\mathcal{X}}(M) = \arg \min_X \|X - M\|_F^2} \quad (108)$$

subject to

$$X \in \mathcal{X}. \quad (109)$$

Here $\mathcal{X}$ is the set of channel matrices satisfying the required rank constraint.

In plain English:

Gradient descent says:

“Move toward a solution that matches the measurements.”

Projection says:

“Now make sure that solution is physically consistent with the multipath structure.”

26 SVD and Low-Rank Projection

For one user, reshape the corresponding row of M into

$$M_k \in \mathbb{C}^{I_1 \times I_2}. \quad (110)$$

Compute its singular value decomposition:

$$\boxed{M_k = U_k \Sigma_k V_k^H} \quad (26)$$

where

$$\Sigma_k = \begin{bmatrix} \sigma_1 & 0 & 0 \\ 0 & \sigma_2 & 0 \\ 0 & 0 & \sigma_3 \end{bmatrix}. \quad (111)$$

The singular values satisfy

$$\sigma_1 \geq \sigma_2 \geq \sigma_3 \geq 0. \quad (112)$$

If the physical channel has only $L_k = 2$ significant paths, retain only the largest two singular values:

$$\tilde{\Sigma}_k = \begin{bmatrix} \sigma_1 & 0 & 0 \\ 0 & \sigma_2 & 0 \\ 0 & 0 & 0 \end{bmatrix}. \quad (113)$$

Then reconstruct:

$$\boxed{\tilde{G}_k = U_k \tilde{\Sigma}_k V_k^H} \quad (27)$$

Now

$$\text{rank}(\tilde{G}_k) = 2. \quad (114)$$

27 Full Numerical Example: PGD Low-Rank Projection

Suppose after one gradient step, the tentative channel matrix for User 1 is

$$M_1 = \begin{bmatrix} 0.81 + 0.24j & -0.36 + 0.71j & 0.92 - 0.41j \\ -0.31 + 0.69j & 0.58 - 0.54j & -0.76 + 0.32j \\ 1.02 + 0.14j & -0.42 - 0.61j & 0.51 + 0.75j \end{bmatrix}. \quad (115)$$

This is a full complex matrix.

It is not diagonal.

27.1 Step 1: Compute SVD

Compute

$$M_1 = U\Sigma V^H. \quad (116)$$

The singular values are

$$\boxed{\sigma_1 = 2.3770, \quad \sigma_2 = 1.0054, \quad \sigma_3 = 0.1116.} \quad (117)$$

Therefore,

$$\Sigma = \begin{bmatrix} 2.3770 & 0 & 0 \\ 0 & 1.0054 & 0 \\ 0 & 0 & 0.1116 \end{bmatrix}. \quad (118)$$

27.2 Step 2: Assume Two Physical Paths

Suppose

$$L_1 = 2. \quad (119)$$

The physical constraint is

$$\text{rank}(G_1) \leq 2. \quad (120)$$

Therefore remove the smallest singular value:

$$\tilde{\Sigma} = \begin{bmatrix} 2.3770 & 0 & 0 \\ 0 & 1.0054 & 0 \\ 0 & 0 & 0 \end{bmatrix}. \quad (121)$$

27.3 Step 3: Reconstruct Rank-2 Matrix

Calculate

$$\tilde{G}_1 = U\tilde{\Sigma}V^H. \quad (122)$$

The projected matrix is

$$\boxed{\tilde{G}_1 = \begin{bmatrix} 0.8115 + 0.2353j & -0.3558 + 0.7231j & 0.9172 - 0.3959j \\ -0.2989 + 0.6721j & 0.5837 - 0.4812j & -0.7862 + 0.3757j \\ 1.0336 + 0.1480j & -0.4640 - 0.6062j & 0.4678 + 0.7313j \end{bmatrix}.} \quad (123)$$

Compare before and after.

Before:

$$\text{rank}(M_1) = 3. \quad (124)$$

After:

$$\boxed{\text{rank}(\tilde{G}_1) = 2.} \tag{125}$$

The numerical values changed slightly, but the matrix now obeys the physical rank constraint. This is exactly the main idea of PGD.

28 PGD Algorithm

Algorithm 1 Projected Gradient Descent for 2D Rydberg Atomic Channel Estimation

Require: $Y_{(3)}, Z_{(3)}, B_{(3)}, S$, ranks $\{L_k\}$, step sizes $\{\zeta_t\}$

```

1: Initialize  $G_{(3)}^{(0)}$ 
2: for  $t = 0, 1, \dots, T - 1$  do
3:   Compute gradient:
      
$$\nabla f(G_{(3)}^{(t)})$$

4:   Take gradient step:
      
$$M = G_{(3)}^{(t)} - \zeta_t \nabla f(G_{(3)}^{(t)})$$

5:   for  $k = 1, \dots, K$  do
6:     Extract row  $[M]_{k,:}$ 
7:     Reshape into:
      
$$M_k \in \mathbb{C}^{I_1 \times I_2}$$

8:     Compute:
      
$$M_k = U_k \Sigma_k V_k^H$$

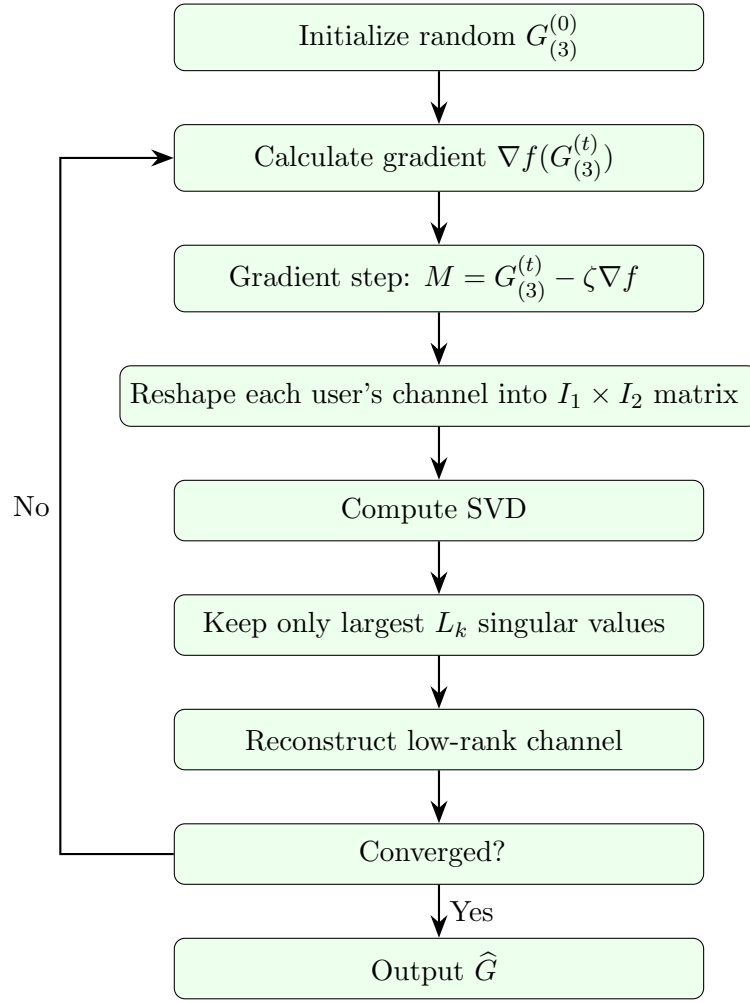
9:     Keep largest  $L_k$  singular values
10:    Set all remaining singular values to zero
11:    Reconstruct:
      
$$\tilde{G}_k = U_k \tilde{\Sigma}_k V_k^H$$

12:    Vectorize  $\tilde{G}_k$ 
13:  end for
14:  Form projected matrix  $X$ 
15:
      
$$G_{(3)}^{(t+1)} = X$$

16:  if  $\left\| G_{(3)}^{(t+1)} - G_{(3)}^{(t)} \right\|_F^2 < \epsilon$  then
17:    Stop
18:  end if
19: end for
20: return  $\hat{G}_{(3)}$ 

```

29 PGD Flowchart



30 Why PGD Can Be Better Than Ordinary GD in 2D

Suppose ordinary GD produces

$$M = \text{measurement-consistent solution.} \quad (126)$$

But measurement consistency alone is not enough.

Noise may create an artificial third singular mode:

$$\sigma_3 > 0. \quad (127)$$

If physics says the user has only two propagation paths, then the third mode is likely not physically meaningful.

PGD removes it:

$$\sigma_3 \rightarrow 0. \quad (128)$$

Thus:

$$\boxed{\text{GD uses measurement information}} \quad (129)$$

while

$$\boxed{\text{PGD uses measurement information + physical structure.}} \quad (130)$$

31 MSE

Suppose the true channel is G and the estimated channel is $\hat{G}$.

The estimation error is

$$E_G = G - \hat{G}. \quad (131)$$

The squared error is

$$\left\| G - \hat{G} \right\|_F^2. \quad (132)$$

The mean square error is

$$\boxed{\text{MSE} = \mathbb{E} \left[\left\| G - \hat{G} \right\|_F^2 \right]}. \quad (133)$$

MSE answers:

“On average, how much squared error exists between the true channel and estimated channel?”

32 NMSE

MSE alone depends on channel magnitude.

Therefore the paper uses normalized MSE:

$$\boxed{\text{NMSE} = \frac{\mathbb{E} \left[\left\| G - \hat{G} \right\|_F^2 \right]}{\mathbb{E} \left[\left\| G \right\|_F^2 \right]}}. \quad (134)$$

For the 2D unfolded model:

$$\boxed{\text{NMSE} = \frac{\mathbb{E} \left[\left\| G_{(3)} - \hat{G}_{(3)} \right\|_F^2 \right]}{\mathbb{E} \left[\left\| G_{(3)} \right\|_F^2 \right]}}. \quad (135)$$

NMSE is often plotted in dB:

$$\boxed{\text{NMSE}_{\text{dB}} = 10 \log_{10}(\text{NMSE})}. \quad (136)$$

Example:

$$\text{NMSE} = 0.01 \quad (137)$$

means

$$\text{NMSE}_{\text{dB}} = 10 \log_{10}(0.01) = -20 \text{ dB}. \quad (138)$$

Lower NMSE means better channel estimation.

33 CRLB

The Cramér–Rao Lower Bound answers:

“Even if we design the best unbiased estimator possible, how small can its estimation error theoretically become?”

It is a theoretical benchmark.

33.1 Vectorized Model

The paper rewrites the 2D model as

$$\boxed{y = \text{Re} \left[z \circ (I \otimes S^T) g \right] + |b| + n} \quad (28)$$

where

$$y = \text{vec}(Y), \quad (139)$$

$$z = \text{vec}(Z), \quad (140)$$

$$g = \text{vec}(G), \quad (141)$$

and $\otimes$ denotes the Kronecker product.

33.2 Probability Density Function

The conditional probability density is written as

$$p(y|g) = \frac{1}{(2\pi\sigma^2)^{I_1 I_2 P}} \exp \left[-\frac{1}{2\sigma^2} \|y - |b| - \text{Re} (z \circ [(I \otimes S^T)g])\|_2^2 \right]. \quad (29)$$

33.3 Fisher Information Matrix

The Fisher information matrix is

$$\boxed{\text{FIM}(g) = -\mathbb{E} \left[\frac{\partial^2 \ln p(y|g)}{\partial g^* \partial g} \right] .} \quad (142)$$

The paper obtains

$$\boxed{\text{FIM}(g) = \frac{1}{4\sigma^4} (I \otimes S^T)^H [\sigma^2 I \circ (z^* z^T)] (I \otimes S^T) .} \quad (30)$$

Since

$$z_i = e^{-j\angle b_i}, \quad (143)$$

the paper simplifies the CRLB to

$$\boxed{\text{CRB}(g) \succeq \text{FIM}(g)^{-1} = 4\sigma^2 [I_{I_1 I_2} \otimes (S^* S^T)^{-1}] .} \quad (31)$$

34 What CRLB Tells Us

Equation (31) gives three important conclusions.

34.1 Higher SNR improves estimation

Higher SNR means smaller noise variance σ^2 .

Therefore:

$$\sigma^2 \downarrow \implies \text{CRLB} \downarrow . \quad (144)$$

So better SNR means a lower possible estimation error.

34.2 More pilots improve estimation

More pilots provide more independent information about the unknown channel.

Therefore increasing P generally improves the inverse pilot-information matrix and lowers estimation uncertainty.

34.3 CRLB is not an algorithm

CRLB does not estimate the channel.

It is only a theoretical lower benchmark.

If an algorithm curve approaches CRLB, that is very good.

35 Complete Comparison: GS, GD and PGD

Feature	GS	GD	PGD
Basic purpose	Recover missing phase	Minimize channel prediction error	Minimize error while enforcing structure
Observation model	Nonlinear magnitude model	Strong-reference approximation	Strong-reference approximation
Main idea	Alternating projections	Follow negative gradient	Gradient + projection
Uses gradient?	No	Yes	Yes
Uses SVD?	No	No	Yes
Works naturally for 1D?	Yes	Yes	GD sufficient
Handles 2D rank constraint?	Not naturally	No explicit rank enforcement	Yes
Physical prior	Magnitude constraint	No explicit low-rank prior	Low-rank multipath prior

36 Numerical Summary of the Three Algorithms

36.1 GD

Input:

$$G^{(0)}, S, Y, B, Z. \quad (145)$$

Compute:

$$R = Y - |B| - \text{Re}[Z \circ (GS)]. \quad (146)$$

Then:

$$\nabla L = -2(R \circ Z^*)S^H. \quad (147)$$

Update:

$$G_{\text{new}} = G_{\text{old}} - \eta \nabla L. \quad (148)$$

36.2 GS

Input:

$$z, S, b. \quad (149)$$

Estimate phase:

$$\theta = \angle(S^H a + b). \quad (150)$$

Restore complex observation:

$$z \circ e^{j\theta}. \quad (151)$$

Remove reference:

$$z \circ e^{j\theta} - b. \quad (152)$$

Least-squares update:

$$a = (SS^H)^{-1}S(z \circ e^{j\theta} - b). \quad (153)$$

36.3 PGD

Gradient step:

$$M = G - \zeta \nabla f(G). \quad (154)$$

SVD:

$$M_k = U_k \Sigma_k V_k^H. \quad (155)$$

Truncate:

$$\sigma_{L_k+1}, \dots \rightarrow 0. \quad (156)$$

Reconstruct:

$$G_k = U_k \tilde{\Sigma}_k V_k^H. \quad (157)$$

37 Complexity

For the 1D GD case, the paper gives complexity

$$\boxed{\mathcal{O}(TPKI)} \quad (158)$$

where

- T is number of iterations,
- P is number of pilots,
- K is number of users,
- I is number of 1D cells.

For 2D PGD, complexity additionally includes the SVD projection:

$$\boxed{\mathcal{O}\left(T \left[K \min(I_1 I_2^2, I_1^2 I_2) + PK I_1 I_2 \right] \right)}. \quad (159)$$

Therefore PGD is computationally heavier than ordinary GD, because SVD must be performed repeatedly.

38 How to Approach Figure 3

Figure 3 studies the 1D antenna array.

The main quantity is

$$\boxed{\text{NMSE versus SNR.}} \quad (160)$$

The paper uses a 1D array with

$$I = 8. \quad (161)$$

The comparison includes:

- GD,
- GS,
- CRLB.

The paper compares different pilot lengths, particularly

$$P = 10 \quad (162)$$

and

$$P = 30. \quad (163)$$

38.1 Simulation procedure

For every SNR point:

1. Generate multipath channel coefficients.
2. Generate complex Gaussian pilots.
3. Generate reference signal.
4. Generate atomic magnitude observations.
5. Run GD.
6. Run GS.
7. Calculate NMSE.
8. Repeat over many random channel/noise realizations.
9. Average NMSE.
10. Calculate CRLB.
11. Plot NMSE versus SNR.

The expected qualitative behavior is:

$$\text{Higher SNR} \Rightarrow \text{Lower NMSE.} \quad (164)$$

With short pilot length, the paper reports GD outperforming GS.

With larger pilot length, GD and GS approach each other and move closer to the CRLB.

39 How to Approach Figure 4

Figure 4 extends the simulation to a 2D atomic array.

The paper uses

$$I_1 = 8, \quad I_2 = 8. \quad (165)$$

Again, the horizontal axis is SNR and vertical axis is NMSE.

The main comparison becomes

- ordinary GD,
- PGD,
- CRLB.

The procedure is

1. Generate 2D multipath channels using Equation (16).
2. Construct tensor $\mathcal{G}$.
3. Generate pilot matrix S .
4. Generate reference tensor $\mathcal{B}$.
5. Generate observations using Equation (19).
6. Unfold tensors along mode 3.
7. Run ordinary GD.
8. Run PGD.
9. For PGD, perform truncated SVD after every gradient step.
10. Calculate NMSE.
11. Average over random realizations.
12. Plot against SNR.

Why should PGD help?

Because when pilots are limited, the observations alone may not contain enough information.

PGD adds another source of knowledge:

$$\boxed{\text{rank}(G_k) \leq L_k.} \quad (166)$$

Thus PGD can reject channel components inconsistent with the assumed multipath geometry.

40 How to Approach Figure 5

Figure 5 studies

$$\boxed{\text{NMSE versus Pilot Length } P.} \quad (167)$$

The SNR is fixed at

$$\text{SNR} = 5 \text{ dB}. \quad (168)$$

The 2D array configuration is used.

The procedure is:

1. Fix SNR to 5 dB.
2. Choose pilot lengths:

$$P = 5, 10, 15, 20, 25, 30, \dots$$

3. For each P , generate a new pilot matrix.
4. Generate observations.
5. Estimate channel using GD.
6. Estimate channel using PGD.
7. Calculate NMSE.
8. Average over random trials.
9. Plot NMSE versus P .

The expected trend is

$$\boxed{P \uparrow \implies \text{NMSE} \downarrow.} \quad (169)$$

More pilots mean more equations containing information about the unknown channel.

The paper reports that performance becomes close to CRLB when the pilot length becomes sufficiently large, around more than 20 pilots in its reported simulation setting.

41 Simulation Parameters Reported in the Paper

The simulation setup includes:

- Number of users:

$$K = 3.$$

- Channel paths:

$$L_k \sim \text{Uniform}(3, 7).$$

- Pilot symbols:

$$s_{k,p} \sim \mathcal{CN}(0, 1).$$

- Channel path gains:

$$\alpha_{l,k} \sim \mathcal{CN}(0, 1).$$

- Reference channel gain:

$$\alpha_b \sim \mathcal{CN}(0, 10).$$

- Polarization components are randomly generated according to the distribution described in the paper.

- PGD initialization:

$$[G_{(3)}^{(0)}]_{k,i} \sim \mathcal{CN}(0, 0.1).$$

42 Recommended Simulation Order

For actually implementing the paper, the recommended learning and coding order is:

Recommended Order

Stage 1: Understand the basic system

Channel $\rightarrow$ Pilots $\rightarrow$ Reference $\rightarrow$ Magnitude observation.

Stage 2: Implement 1D GD

$G, S, B, Z, Y \rightarrow$ Loss $\rightarrow$ Gradient $\rightarrow$ GD.

Stage 3: Implement GS baseline

Magnitude $\rightarrow$ Phase estimate $\rightarrow$ LS projection $\rightarrow$ repeat.

Stage 4: Verify NMSE

Before plotting anything, verify:

$$\hat{G} \approx G_{\text{true}}$$

at high SNR.

Stage 5: Generate Fig. 3

1D:

NMSE versus SNR.

Stage 6: Extend to 2D

Generate:

$$\mathcal{G} \in \mathbb{C}^{I_1 \times I_2 \times K}.$$

Stage 7: Implement PGD

Gradient $\rightarrow$ Reshape $\rightarrow$ SVD $\rightarrow$ Truncate $\rightarrow$ Reconstruct.

Stage 8: Generate Fig. 4

2D:

NMSE versus SNR.

Stage 9: Generate Fig. 5

2D:

NMSE versus Pilot Length.

43 Important Assumptions

The mathematical framework relies on several assumptions.

43.1 Strong reference assumption

$$|b_{i,p}| \gg |a_{i,p} + n_{i,p}|. \quad (170)$$

Without this condition, the Taylor approximation may become less accurate.

43.2 Perfect synchronization

The paper assumes synchronization and prior coordination for pilot transmission.

43.3 Known reference

The reference signal characteristics are assumed known or obtainable.

43.4 Multipath low-rank structure

For PGD:

$$\text{rank}(G_k) \leq L_k. \quad (171)$$

This physical prior is crucial.

43.5 Known or estimated rank

PGD requires the path number L_k or an appropriate rank choice.

44 Limitations

1. If the reference is not sufficiently stronger than the signal, the strong-reference approximation can deteriorate.
2. GS is naturally designed around magnitude constraints and does not naturally enforce the 2D low-rank channel structure.
3. Ordinary GD does not explicitly use the multipath rank constraint.
4. PGD requires repeated SVD operations, increasing computational complexity.
5. Incorrect selection of L_k may cause underfitting or overfitting.
6. CRLB is a lower bound and is not automatically achieved by finite-iteration practical algorithms.
7. Reproducing simulation curves requires matching random distributions, initialization, step sizes, stopping criteria and Monte-Carlo averaging.

45 Master Comparison Example

The three algorithms can be remembered using one problem.

Suppose the receiver measures

$$y = |Gs + b + n|. \quad (172)$$

45.1 GS

GS says:

“I know the magnitude. Let me estimate the missing phase, then solve the resulting linear problem.”

45.2 GD

GD says:

“Because the reference is strong, I can approximately linearize the measurement. Let me iteratively reduce the prediction error.”

45.3 PGD

PGD says:

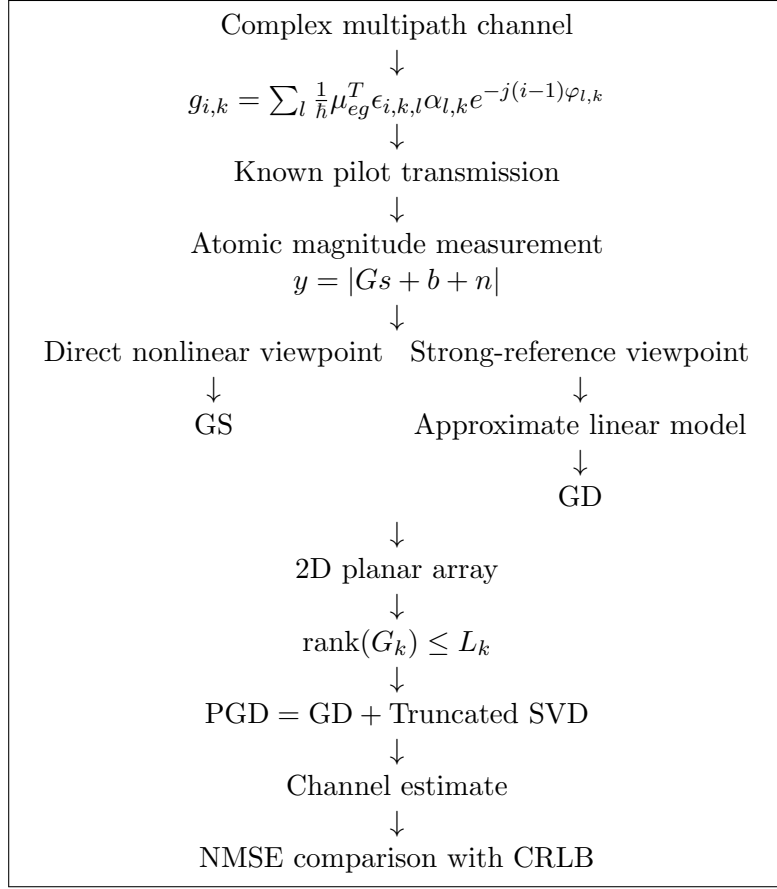
“I will reduce the prediction error, but after every update I will force the channel to obey the low-rank multipath structure.”

Therefore:

GS	→	Phase recovery
GD	→	Error minimization
PGD	→	Error minimization + Physical projection

(173)

46 Final Concept Map



(174)

47 Final Takeaway

The entire paper can be summarized by the following statement:

Core Understanding

A Rydberg atomic receiver does not directly provide the usual complex linear observation used in conventional channel estimation. Instead, its atomic response creates a nonlinear magnitude-type measurement. A strong known reference approximately converts this difficult nonlinear observation into a real-valued optimization model.

For a 1D atomic array, the channel can be estimated using gradient descent, while biased Gerchberg–Saxton provides a phase-retrieval baseline.

For a 2D atomic array, each user’s spatial channel becomes a matrix. Because a physical multipath channel contains only a limited number of paths, this matrix has a low-rank structure. Projected Gradient Descent exploits this structure by taking a normal gradient step and then using truncated SVD to remove channel components inconsistent with the physical multipath model.

Finally, NMSE measures estimation quality and CRLB provides the theoretical benchmark for the best possible unbiased estimator.